Functions & Transformations Maze **ANSWER KEY**

Name _____ Period _____

Start in the top-left room. Solve each problem. Go through the door that shows your answer. Show your work on the answer grid.

START

1 Let $f(x) = 3x^2 - 4x + 1$.
Find $f(-2)$.

21 | $x > 3$ | $(-\infty, -1) \cup (-1, \infty)$

2 State the domain.
$f(x) = \sqrt{(2x - 6)}$

$x \geq 6$ | **$x \geq 3$** | $(2, -3)$

$f(x) = x^2 - 3x$
and $g(x) = 2x + 1$.
Find $(f \circ g)(3)$.

8 | $3x + 4$ | -13

6 $f(x) = \begin{cases} x^2 - 1, & x < 2 \\ 3x - 5, & x \geq 2 \end{cases}$
Find $f(-3) + f(2)$.

$3x - 4$ | **9** | 3

State the domain in interval notation.
$f(x) = \dfrac{x + 1}{x - 3}$

$(-\infty, 3]$ | $|x + 3| + 2$ | $(6, -3)$

3 The point $(4, -3)$ is on $y = f(x)$.
Find its image on $y = -f(x + 2)$.

$3x^3 - 2x^2 + 3x - 2$ | **$(2, 3)$** | **$9x^2 - 12x + 5$**

5 Find the inverse.
$f(x) = \dfrac{x + 4}{3}$

$(3, \infty)$ | -13 | 4

7 Find $(f \circ g)(2)$.

x	1	2	3
$f(x)$	3	1	4
$g(x)$	2	3	1

3 | $y \leq -5$ | $3\sqrt{(4 - x)}$

Write the equation of $y = |x|$ after it shifts left 3 and down 2.

$9x^2 + 5$ | $|x - 3| - 2$ | $3x^2 + 1$

4 $f(x) = 3x - 2$
and $g(x) = x^2 + 1$.
Find $(g \circ f)(x)$.

$x \leq 3$ | 2 | **$(x - 1)^2 + 2, x \geq 1$**

8 Find the inverse, with its domain.
$f(x) = \sqrt{(x - 2)} + 1$

$(x - 1)^2 - 2, x \geq 1$ | 5 | $3\sqrt{(4 - x)}$

State the range.
$f(x) = -2(x - 1)^2 + 5$

$y \leq -5$

The point $(6, -4)$ is on $y = f(x)$.
Find its image on $y = 3f(2x) - 1$.

$64

9 A \$80 jacket: first a \$15 coupon, then 20% off the reduced price.
What is the final price?

$52

10 Write the equation of $y = \sqrt{x}$ after a vertical stretch by 3, a reflection in the x-axis, and a shift right 4.

$-3\sqrt{(x - 4)}$

FINISH

You made it!

Functions Maze: Answer Grid **ANSWER KEY**

Name _____ Period _____

Write each answer in the order you found it, and show your work for that room.

1	Answer: **21**	$3(-2)^2 - 4(-2) + 1 = 12 + 8 + 1 \rightarrow 21$
2	Answer: **$x \geq 3$**	$2x - 6 \geq 0 \rightarrow 2x \geq 6 \rightarrow x \geq 3 \rightarrow x \geq 3$
3	Answer: **(2, 3)**	x: shift LEFT 2 $\rightarrow 4 - 2 = 2$; reflect in the x-axis: $-(-3) = 3 \rightarrow (2, 3)$
4	Answer: **$9x^2 - 12x + 5$**	$g(3x - 2) = (3x - 2)^2 + 1 \rightarrow 9x^2 - 12x + 5$
5	Answer: **$3x - 4$**	$y = \frac{x + 4}{3} \rightarrow$ swap: $x = \frac{y + 4}{3} \rightarrow y = 3x - 4 \rightarrow 3x - 4$
6	Answer: **9**	$x = -3 < 2$: $9 - 1 = 8$; $x = 2$ (≥ 2): $3(2) - 5 = 1 \rightarrow 9$
7	Answer: **4**	$g(2) = 3$, then $f(3) = 4 \rightarrow 4$
8	Answer: **$(x - 1)^2 + 2, x \geq 1$**	swap: $x = \sqrt{(y - 2)} + 1 \rightarrow y = (x - 1)^2 + 2$; range of f is $y \geq 1 \rightarrow (x - 1)^2 + 2, x \geq 1$
9	Answer: **$52**	$80 - 15 = 65$, then $0.80(65) \rightarrow \$52$
10	Answer: **$-3\sqrt{(x - 4)}$**	$3\sqrt{x} \rightarrow -3\sqrt{x} \rightarrow$ replace x with $x - 4 \rightarrow -3\sqrt{(x - 4)}$